Section 1. Number Conversions.

Direction: Read the questions and do the required conversions by showing the steps. (2 x 2= 4 Marks)

1) Convert the following Binary to its Decimal form. (2 Marks)

64 32 16 8 4 2 1

1001011 (2) =? (75)(10)

$1 + 2 + 8 + 64 = 75_{10}$

2

2) Convert the following Hexadecimal to its Binary form. (2 Marks)

56E (16) =? (0011 0110 0001 0100) (2)

14

0.5

56 14

(8 4 21)

4 = 0100

1 = 0001

6 = 0110

5 = 0011

$(0011\,0110\,0001\,0100)_2$

Section 2. Binary Arithmetic. (1 Mark)

1) Calculate the addition for the following two (2) binary numbers by showing the steps. (1 Mark)

11111 + 11010 =?

1

Answer:

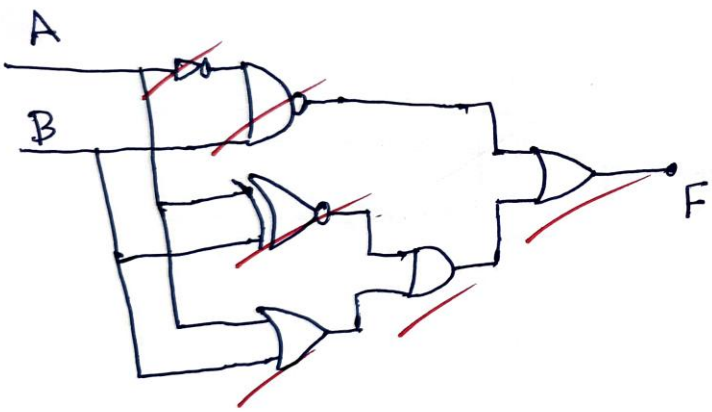
```
 1 1 1
 11111
 11010 +
 ------
 111001
```

Section 3. Logic Gates. (5 marks)

1) Draw the logic gate of the following expression. (2 Marks)

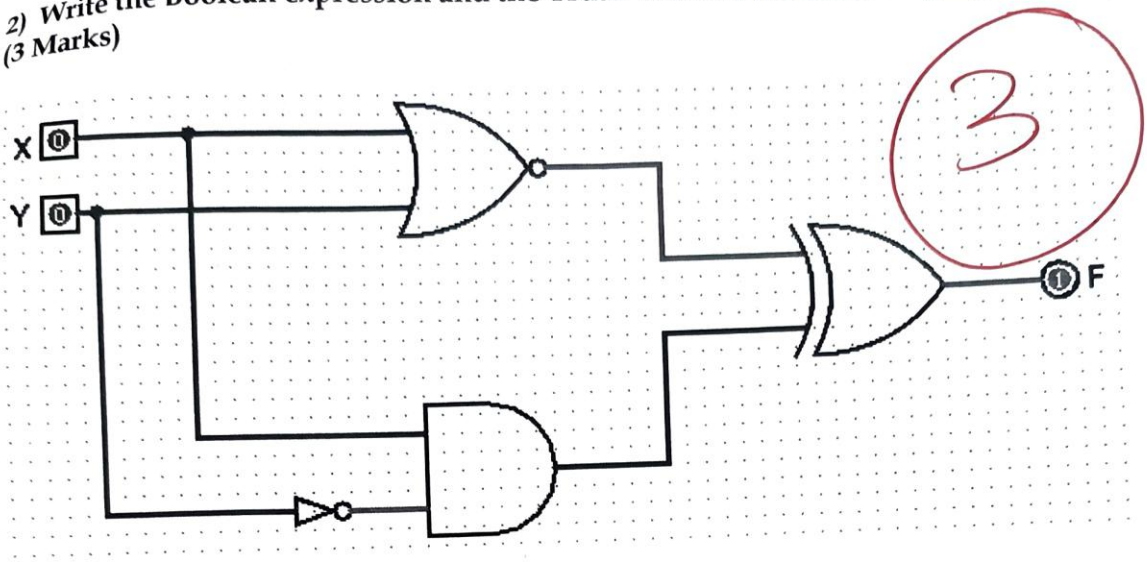
$F = (A'B)' + [A \oplus B)' (A+B)]$

Answer:



2

2) Write the Boolean expression and the Truth Table of the following logic circuit. (3 Marks)



3

Answer:

Boolean Expression:

$F = [(X+Y)' \oplus (Xy')]$ 1

Truth Table:

X	Y	$(X+y)'$	y'	(Xy')	F
0	0	1	1	0	1
0	1	0	0	0	0
1	0	0	1	1	1
1	1	0	0	0	0

2

**** End of Exam ****